Test Plan Document – Web Based ML Model Evaluator

Purpose: This test plan covers the functional requirements of the Web Based ML Model Evaluator. Eight test cases are provided for each functional requirement, including Unit, Integration, and System test cases. Actual Result and Test Result are intentionally left blank for manual execution.

Test case convention: FR01-UT-01 = Functional Requirement 1, Unit Test, Test Case 01; similarly for IT and ST.

Total test cases: 80 (8 test cases for each of the 10 functional requirements).

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
FR01-UT-01	Dataset Upload	Verify that the upload component accepts a valid CSV file.	Application is running and upload component is available.	1. Open dataset upload page. 2. Select a CSV file. 3. Submit the file.	students.csv; 20 rows, 5 columns.	The file is accepted and upload processing starts.		
FR01-UT-02	Dataset Upload	Verify rejection of a non-CSV file.	Upload component is available.	1. Select a non-CSV file. 2. Submit the file.	students.xlsx	The system rejects the file and displays a clear unsupported-file-type message.		
FR01-UT-03	Dataset Upload	Verify that uploaded CSV metadata is calculated.	Application is running.	1. Upload a valid CSV. 2. Wait for upload processing.	students.csv; 20 rows, 5 columns.	The system reports 20 rows and 5 columns.		
FR01-IT-01	Dataset Upload	Verify frontend-to-backend transfer of a CSV upload.	User is logged in and upload page is open.	1. Choose valid CSV. 2. Click Upload. 3. Observe request/response and UI.	classification.csv; 100 rows, 6 columns.	Backend receives the file, processes it, and returns a successful response; UI shows upload success.		
FR01-IT-02	Dataset Upload	Verify backend handling of an empty CSV.	User is logged in.	1. Select empty.csv. 2. Upload it.	empty.csv; 0 rows.	Backend rejects the file with a meaningful validation error and the UI does not crash.		
FR01-IT-03	Dataset Upload	Verify uploaded file metadata is stored for the user session.	User is logged in and database is available.	1. Upload students.csv. 2. Open the upload/dashboard record.	students.csv; 20 rows, 5 columns.	The upload record and its metadata are associated with the current user only.		
FR01-ST-01	Dataset Upload	Verify end-to-end valid dataset upload.	User has an account and is logged in.	1. Navigate to upload. 2. Select students.csv. 3. Upload. 4. Continue to the next step.	students.csv; valid tabular CSV.	Dataset uploads successfully and the user can proceed to preview/preprocessing.		
FR01-ST-02	Dataset Upload	Verify end-to-end handling of a corrupted CSV.	User is logged in.	1. Select corrupted.csv. 2. Upload.	corrupted.csv; malformed CSV structure.	The application displays a clear error and remains usable without a blank page or crash.		
FR02-UT-01	Dataset Preview	Verify preview displays the first rows of an uploaded dataset.	A valid CSV has been uploaded.	1. Open dataset preview.	students.csv; 20 rows.	A table containing the first few dataset rows is displayed.		
FR02-UT-02	Dataset Preview	Verify column names are displayed in preview.	Valid dataset is available.	1. Open preview.	students.csv columns: Student_ID, Age, Gender, Attendance, Result.	All dataset column names are displayed correctly.		
FR02-UT-03	Dataset Preview	Verify preview handles a dataset with exactly 5 columns.	Valid CSV is available.	1. Upload/open preview.	dataset5.csv; 20 rows, 5 columns.	Five columns are displayed with correct values.		
FR02-IT-01	Dataset Preview	Verify uploaded data is passed correctly from backend to preview UI.	Valid CSV has been uploaded.	1. Upload dataset. 2. Request preview. 3. Compare UI values with source CSV.	First 5 rows of students.csv.	Preview data returned by backend matches the source CSV and is rendered correctly.		

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FR02-IT-02	Dataset Preview	Verify preview uses the current user's uploaded dataset.	Two users have separate uploaded datasets.	1. User A uploads A.csv. 2. User B uploads B.csv. 3. Each opens preview.	A.csv contains marker USER_A; B.csv contains USER_B.	Each user sees only their own dataset preview.		
FR02-IT-03	Dataset Preview	Verify preview returns a controlled error when dataset parsing fails.	Malformed dataset is available.	1. Attempt to open preview for malformed.csv.	malformed.csv; inconsistent row lengths.	Backend returns an error response and the UI displays a meaningful message.		
FR02-ST-01	Dataset Preview	Verify end-to-end dataset inspection before target selection.	Valid CSV has been uploaded.	1. Upload dataset. 2. Open preview. 3. Inspect columns and sample values.	students.csv; 20 rows, 5 columns.	The user can inspect column names and sample values before selecting the target.		
FR02-ST-02	Dataset Preview	Verify preview remains usable for a larger reasonable CSV.	Application is running.	1. Upload a larger valid CSV. 2. Open preview.	large_students.csv; 5,000 rows, 8 columns.	Preview loads successfully without requiring the entire dataset to be rendered at once.		
FR03-UT-01	Data Preprocessing	Verify missing-value filling option replaces null numeric values.	Preprocessing component is available.	1. Load dataset. 2. Select fill-missing option. 3. Apply.	Age column: [20, 21, null, 23]. Fill with mean.	Null value is replaced according to the selected fill strategy.		
FR03-UT-02	Data Preprocessing	Verify categorical encoding converts text values to numeric representation.	Preprocessing component is available.	1. Select categorical column. 2. Choose one-hot encoding. 3. Apply.	Gender: [Male, Female, Female].	Categorical values are encoded into numeric columns.		
FR03-UT-03	Data Preprocessing	Verify drop-null option removes rows containing missing values.	Preprocessing component is available.	1. Select drop-null option. 2. Apply.	Age: [20, null, 22].	Rows containing null values are removed.		
FR03-IT-01	Data Preprocessing	Verify preprocessing selections are sent from UI to backend.	Dataset is uploaded and preprocessing page is open.	1. Choose fill-null and one-hot encoding. 2. Apply preprocessing.	Age has 2 nulls; Gender is categorical.	Backend receives selected preprocessing options and returns the processed dataset.		
FR03-IT-02	Data Preprocessing	Verify preprocessing output can be passed to model-training logic.	Processed dataset is available.	1. Apply preprocessing. 2. Select target. 3. Start training.	Dataset with nulls and Gender text column.	Training receives cleaned numeric features without preprocessing-related type/null errors.		
FR03-IT-03	Data Preprocessing	Verify invalid preprocessing configuration is rejected.	Preprocessing endpoint is available.	1. Submit an invalid/unsupported preprocessing option.	preprocess_option = UNKNOWN	Backend returns a validation error and does not corrupt the dataset.		
FR03-ST-01	Data Preprocessing	Verify end-to-end preprocessing for a classification dataset.	Valid CSV uploaded.	1. Choose target Result. 2. Fill missing values. 3. Encode Gender. 4. Continue.	classification.csv with missing Age values and Gender column.	Preprocessing completes and the dataset is ready for classification training.		
FR03-ST-02	Data Preprocessing	Verify user can proceed after preprocessing without re-uploading.	Valid CSV uploaded and preprocessing completed.	1. Apply preprocessing. 2. Navigate to model selection. 3. Return to preprocessing and change an option.	students.csv; missing values present.	The user can modify preprocessing settings without having to upload the same dataset again.		
FR04-UT-01	Target Column Selection	Verify a valid dataset column can be selected as target.	Dataset preview is available.	1. Open target selection. 2. Select Result.	Result is a column in students.csv.	Result is accepted as the target column.		
FR04-UT-02	Target Column Selection	Verify target selection cannot be empty.	Dataset is uploaded.	1. Leave target selection blank. 2. Click Continue.	No target selected.	System prevents continuation and displays a clear target-selection message.		
FR04-UT-03	Target Column Selection	Verify selected target is stored as the output variable.	Dataset is uploaded.	1. Select Result. 2. Save/continue. 3. Inspect selected configuration.	Target = Result.	Configuration records Result as the target and remaining suitable columns as features.		
FR04-IT-01	Target Column Selection	Verify selected target is sent from frontend to backend.	Dataset and target-selection page are available.	1. Select Result. 2. Continue to training configuration.	target_column = Result.	Backend receives target_column=Result and returns success.		
FR04-IT-02	Target Column Selection	Verify invalid target column sent to backend is rejected.	Backend target endpoint is available.	1. Submit a target name not present in the dataset.	target_column = DoesNotExist	Backend returns a validation error and training cannot start.		

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FR04-IT-03	Target Column Selection	Verify target selection controls feature/target separation.	Dataset has multiple columns.	1. Select Result as target. 2. Prepare training data.	Features: Student_ID, Age, Gender, Attendance; Target: Result.	Result is excluded from input features and used as the output variable.		
FR04-ST-01	Target Column Selection	Verify end-to-end target selection before training.	Valid dataset is uploaded.	1. Preview dataset. 2. Select Result. 3. Continue.	students.csv; target Result.	User can proceed to split/model selection with the correct target.		
FR04-ST-02	Target Column Selection	Verify changing the target updates the training configuration.	Dataset contains two eligible target-like columns.	1. Select Result. 2. Change selection to another valid column. 3. Continue.	dataset.csv; targets Result and Grade.	The latest selected column is used as target and the previous selection is removed.		
FR05-UT-01	Train-Test Split	Verify a 70-30 split produces the expected partition.	Dataset is loaded and target is selected.	1. Set split to 70-30. 2. Apply split.	100-row dataset.	Approximately 70 rows are used for training and 30 for testing.		
FR05-UT-02	Train-Test Split	Verify an 80-20 split produces the expected partition.	Dataset is loaded.	1. Set split to 80-20. 2. Apply split.	100-row dataset.	Approximately 80 rows are used for training and 20 for testing.		
FR05-UT-03	Train-Test Split	Verify invalid split values are rejected.	Split control is available.	1. Enter an invalid ratio. 2. Apply.	Train=0%, Test=100% or Train=110%, Test=-10%.	Invalid split is rejected with a validation message.		
FR05-IT-01	Train-Test Split	Verify selected split ratio reaches training backend.	Target is selected.	1. Select 70-30. 2. Start training.	split_ratio = 0.70.	Backend receives and applies the requested split ratio.		
FR05-IT-02	Train-Test Split	Verify test data is kept separate from training data.	Dataset and split logic are available.	1. Split a dataset. 2. Inspect training and test records.	100-row dataset; 80-20 split.	No test record is included in the training partition.		
FR05-IT-03	Train-Test Split	Verify split and target data are aligned.	Dataset target is selected.	1. Apply 80-20 split. 2. Compare X_train/y_train and X_test/y_test sizes.	100 rows; target Result.	Feature and target partitions have matching row counts.		
FR05-ST-01	Train-Test Split	Verify user can retrain with a different split without re-uploading.	Dataset is already uploaded.	1. Train with 70-30. 2. Change to 80-20. 3. Retrain.	students.csv; 70-30 then 80-20.	Second training run uses 80-20 and does not require re-uploading the dataset.		
FR05-ST-02	Train-Test Split	Verify training is not evaluated on training data.	Dataset is uploaded and model selected.	1. Select split. 2. Train model. 3. View evaluation results.	students.csv; 80-20 split.	Evaluation metrics are calculated on the test partition, not the training partition.		
FR06-UT-01	ML Algorithm Selection	Verify supported classification algorithms are listed.	Classification target is selected.	1. Open model selection dropdown.	Classification dataset.	Supported classification models such as Logistic Regression, Decision Tree, Random Forest and KNN are available.		
FR06-UT-02	ML Algorithm Selection	Verify a supported regression algorithm can be selected for regression.	Regression target is selected.	1. Open model selection.	House-price dataset; target Price.	Supported regression model options are available.		
FR06-UT-03	ML Algorithm Selection	Verify an algorithm can be selected without code.	Model selection page is open.	1. Open dropdown. 2. Select Random Forest.	Random Forest.	Random Forest becomes the selected model.		
FR06-IT-01	ML Algorithm Selection	Verify selected classification algorithm is sent to backend.	Classification dataset and target are configured.	1. Select Decision Tree. 2. Start training.	algorithm = Decision Tree.	Backend receives the selected algorithm and initializes the corresponding model.		
FR06-IT-02	ML Algorithm Selection	Verify incompatible algorithm/problem configuration is handled.	Regression target is selected.	1. Choose a classification-only algorithm if unavailable/incompatible. 2. Continue.	Regression dataset; classification algorithm.	System prevents invalid selection or displays a clear compatibility error.		
FR06-IT-03	ML Algorithm Selection	Verify unsupported algorithm value sent to backend is rejected.	Backend endpoint is available.	1. Submit an unsupported algorithm name.	algorithm = UnsupportedModel	Backend returns a validation error and does not start training.		
FR06-ST-01	ML Algorithm Selection	Verify end-to-end selection and training of a classification algorithm.	Classification dataset is uploaded and target selected.	1. Select Random Forest. 2. Set split. 3. Train.	classification.csv; target Result; Random Forest.	Selected algorithm trains successfully and results are displayed.		

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FR06-ST-02	ML Algorithm Selection	Verify end-to-end selection and training of a regression algorithm.	Regression dataset is uploaded and target selected.	1. Select a supported regression algorithm. 2. Set split. 3. Train.	housing.csv; target Price.	Regression model trains successfully and regression evaluation results are produced.		
FR07-UT-01	Model Training	Verify training starts for a valid model configuration.	Preprocessed dataset, target, split and model are configured.	1. Click Train.	classification.csv; Random Forest; 80-20.	Training process starts without validation errors.		
FR07-UT-02	Model Training	Verify training completion state is generated.	Valid training configuration is available.	1. Start training. 2. Wait for completion.	students.csv; Decision Tree.	Training completes and a success/completion state is generated.		
FR07-UT-03	Model Training	Verify training failure is reported when input data is invalid.	Invalid training data is prepared.	1. Start training.	Dataset with non-numeric unprocessed feature and preprocessing disabled.	Training fails gracefully with a clear error rather than crashing.		
FR07-IT-01	Model Training	Verify model-training request reaches the ML backend.	Frontend and backend are running.	1. Configure model. 2. Click Train. 3. Observe response.	algorithm=Logistic Regression; split=80-20.	Backend receives the request and returns training status/results.		
FR07-IT-02	Model Training	Verify trained model is stored after successful training.	Database/storage is available.	1. Train a valid model. 2. Check model storage/metadata.	Random Forest trained on classification.csv.	A trained model artifact and relevant metadata are stored for later reuse.		
FR07-IT-03	Model Training	Verify training status is reflected by the frontend.	Training endpoint is available.	1. Start training. 2. Observe UI while request is running and after completion.	Valid medium-size dataset.	UI indicates training is in progress and then shows success/failure status.		
FR07-ST-01	Model Training	Verify complete training workflow from dataset to trained model.	User is logged in.	1. Upload CSV. 2. Preview. 3. Preprocess. 4. Select target. 5. Set split. 6. Select model. 7. Train.	students.csv; target Result; 80-20; Random Forest.	The selected model is trained successfully and a trained-model result is available.		
FR07-ST-02	Model Training	Verify application remains usable after training failure.	User is logged in.	1. Configure invalid training input. 2. Start training. 3. Read error. 4. Return to configuration.	Invalid/incompatible dataset configuration.	Clear failure feedback is shown and the application remains usable for correction and retry.		
FR08-UT-01	Model Evaluation	Verify classification accuracy is calculated.	A classification model has predictions and true labels.	1. Run evaluation.	y_test and predictions with 8 correct out of 10.	Accuracy is calculated as 0.80 or 80%.		
FR08-UT-02	Model Evaluation	Verify classification precision, recall and F1-score are produced.	Classification predictions are available.	1. Run evaluation.	Binary classification y_test/predictions.	Precision, recall and F1-score are calculated and displayed.		
FR08-UT-03	Model Evaluation	Verify regression RMSE and R2 are calculated.	Regression model predictions are available.	1. Run evaluation.	y_test=[10,20,30]; predictions=[12,18,33].	RMSE and R2 are calculated and displayed for the regression model.		
FR08-IT-01	Model Evaluation	Verify classification model output feeds the correct metrics.	Classification model is trained.	1. Complete training. 2. Open evaluation.	Logistic Regression on classification.csv.	Evaluation uses model predictions and displays classification metrics.		
FR08-IT-02	Model Evaluation	Verify regression model output feeds regression metrics.	Regression model is trained.	1. Complete training. 2. Open evaluation.	Regression model on housing.csv.	Evaluation displays RMSE and R2 rather than classification-only metrics.		
FR08-IT-03	Model Evaluation	Verify confusion matrix is generated for classification.	Classification evaluation is available.	1. Open classification results.	Binary target; y_test and predictions available.	A confusion matrix is generated as part of the classification evaluation output.		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
FR08-ST-01	Model Evaluation	Verify end-to-end evaluation results are understandable.	Model training has completed.	1. Open results page. 2. Inspect metric table/charts.	Classification model with valid test set.	Accuracy, precision, recall and F1-score are shown in an easy-to-read format with relevant visualization.		
FR08-ST-02	Model Evaluation	Verify evaluation is performed on unseen test data.	Model trained using a train-test split.	1. Complete training. 2. View metrics. 3. Confirm evaluation partition.	80-20 split.	Displayed metrics correspond to the held-out test set and not the training data.		
FR09-UT-01	Model Comparison	Verify two trained models can be selected for comparison.	Two models have completed training on the same dataset/split.	1. Open comparison. 2. Select Model A and Model B.	Logistic Regression and Decision Tree.	Both models are accepted for side-by-side comparison.		
FR09-UT-02	Model Comparison	Verify comparison displays the same metric categories for comparable models.	Two classification models are available.	1. Select both models. 2. Compare results.	Random Forest and KNN; same classification dataset.	Relevant common metrics are displayed for both models.		
FR09-UT-03	Model Comparison	Verify duplicate selection is handled.	One trained model is available.	1. Select the same model twice.	Random Forest selected twice.	System prevents duplicate comparison or clearly indicates that two distinct models are required.		
FR09-IT-01	Model Comparison	Verify comparison request retrieves results for selected models.	Model result records exist.	1. Select two models. 2. Submit comparison.	Model IDs M01 and M02.	Backend returns evaluation results for both selected models.		
FR09-IT-02	Model Comparison	Verify comparison uses the same dataset and split.	Two model results are available.	1. Select results trained on same dataset/split. 2. Compare.	students.csv; 80-20; Random Forest vs KNN.	Comparison is permitted and results are directly comparable.		
FR09-IT-03	Model Comparison	Verify incompatible comparison is rejected or clearly qualified.	Two results use different datasets/splits.	1. Select both results. 2. Compare.	Model A: students.csv 80-20; Model B: housing.csv 70-30.	System prevents misleading comparison or clearly reports that the configurations differ.		
FR09-ST-01	Model Comparison	Verify end-to-end comparison of at least two algorithms.	Dataset is uploaded and target selected.	1. Train Logistic Regression. 2. Train Decision Tree. 3. Open comparison. 4. Select both.	classification.csv; same 80-20 split.	Both model results appear side by side with comparable metrics.		
FR09-ST-02	Model Comparison	Verify visual comparison helps identify the better-performing model.	Two trained models have results.	1. Open comparison. 2. Inspect metric bar chart/table.	Random Forest and KNN results.	Charts/table clearly show metric values for each model so the user can compare performance.		
FR10-UT-01	Model and Report Download	Verify trained model download control is available after training.	A model has trained successfully.	1. Open model results.	Trained Random Forest model.	A model download option is displayed.		
FR10-UT-02	Model and Report Download	Verify evaluation report download control is available.	Evaluation results are available.	1. Open results. 2. Select report download.	Classification evaluation with metrics.	A report download option is available.		
FR10-UT-03	Model and Report Download	Verify downloaded report contains required evaluation information.	Report generation is available.	1. Generate/download report. 2. Open the generated PDF.	Model=Random Forest; metrics=accuracy, precision, recall, F1.	PDF contains the model used and evaluation metrics.		
FR10-IT-01	Model and Report Download	Verify backend model artifact is returned to download endpoint.	A trained model is stored.	1. Request model download. 2. Save returned file.	Stored model M01.	Backend returns the correct trained model artifact for download.		
FR10-IT-02	Model and Report Download	Verify report generation uses stored evaluation results.	Evaluation record exists.	1. Request report generation.	Model M01; stored accuracy/precision/recall/F1.	Generated report contains the stored results and model information.		
FR10-IT-03	Model and Report Download	Verify download error is handled for a missing model/report.	Download endpoint is available.	1. Request a non-existent model/report.	model_id = INVALID123	System returns a controlled not-found/error response and does not crash.		

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
FR10-ST-01	Model and Report Download	Verify end-to-end trained model download.	A model has been trained successfully.	1. Train model. 2. Open results. 3. Click Download Model. 4. Verify downloaded file.	Random Forest trained on classification.csv.	A valid trained model file is downloaded and can be identified as the selected model.		
FR10-ST-02	Model and Report Download	Verify end-to-end evaluation report download.	Evaluation is complete.	1. Train model. 2. Open evaluation. 3. Download PDF report. 4. Open report.	Classification model; accuracy, precision, recall and F1-score.	PDF downloads successfully and summarizes the selected model and evaluation results.		

Manual execution note: The Actual Result and Test Result columns must be populated after executing each test case. Do not mark Pass/Fail until the actual behavior has been observed.